Introductory Finance Final Review

Finance 526

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Outline

- The basic framework of valuation/asset pricing:

$$PV_0 = \sum_{t=0}^{\infty} \frac{CF_t}{(1+r)^t}$$

- CF: Free cash flows in capital budgeting, Dividends in stock valuation, Coupon and principal payments in bond valuation.
- r: The discount rate/ required (expected) return/ cost of capital

r = risk-free rate + risk premium

- In Module 1&2&3, we learn FCF and the stock dividend discount model, assuming that r is constant.
- In Module 4, we use Treasury bonds to learn about the risk-free rate.
- In Module 5&6&7, we use CAPM to learn about the risk premium.
- In Module 8&9, we combine the risk in equity and debt to compute (unlevered/levered) WACC.

Module 1: PV Formulas

- **Constant perpetuity**:

$$PV_0 = \frac{C_1}{1+r} + \frac{C_1}{(1+r)^2} + \frac{C_1}{(1+r)^3} + \cdots = \frac{C_1}{r}$$

- **Growing perpetuity:**

$$PV_0 = \frac{C_1}{1+r} + \frac{C_1(1+g)}{(1+r)^2} + \frac{C_1(1+g)^2}{(1+r)^3} + \cdots = \frac{C_1}{r-g}$$

- **Constant annuity** paying C at the end of each of n periods

$$PV_0 = \frac{C_1}{1+r} + \frac{C_1}{(1+r)^2} + \cdots + \frac{C_1}{(1+r)^n} = \frac{C_1}{r}\left(1 - \frac{1}{(1+r)^n}\right)$$

- **Growing annuity** with n periods:

$$PV_0 = \frac{C_1}{1+r} + \frac{C_1(1+g)}{(1+r)^2} + \cdots + \frac{C_1(1+g)^{n-1}}{(1+r)^n} = \frac{C_1}{r-g}\left(1 - \left(\frac{1+g}{1+r}\right)^n\right)$$

Pay attention to the timing of cash flows!

Module 2: FCF, NPV, IRR

- **FCF**:

$$FCF = \overbrace{(\text{Revenues} - \text{Costs} - \text{Depreciation}) \times (1 - \tau_c)}^{\text{Unlevered Net Income}}$$
$$+ \text{Depreciation} - \text{CapEx} - \Delta\text{NWC}$$

- **NPV**:

$$NPV = C_0 + \sum_{t=1}^{T} \frac{C_t}{(1+r)^t}$$

- **IRR**:

$$NPV = \sum_{t=0}^{T} \frac{C_t}{(1+IRR)^t} = 0$$

Module 3: Stock Valuation Models

- **Dividend discount model**, i.e. the price for a stock, P_0, equals to the present value of all discounted future dividends:

$$\boxed{P_0 = \sum_{t=1}^{\infty} \frac{Div_t}{(1+r_E)^t}}$$

- Gordon Growth Model:

$$P_0 = \frac{Div_1}{r_E - g}$$

- PV of growth opportunities:

$$P_0 = P_0^{NG} + PV(Growth\ Opp) = \frac{EPS_1}{r_E} + PV(Growth\ Opp)$$

$$\Rightarrow PV(Growth\ Opp) = P_0 - \frac{EPS_1}{r_E}$$

Module 4: Bond Valuation

$$P = \sum_{t=1}^{T} \frac{C_t}{(1+r_t)^t} + \frac{F}{(1+r_T)^T}$$

- Compute spot rates:
  - For a ZCB: $P = \frac{F}{(1+r_t)^t} \quad \Rightarrow \quad r_t = \left(\frac{F}{P}\right)^{1/t} - 1$
  - For a coupon bond: Use multiple coupon bonds to solve system of equations.
- Compute forward rates:

$$(1+r_t)^t = (1+r_{t-1})^{t-1}(1+f_t)$$

- Compute YTM:
  - For a ZCB, YTM = spot rate.
  - For a coupon bond, $P = \sum_{t=1}^{T} \frac{C_t}{(1+y)^t} + \frac{F}{(1+y)^T} \quad \Rightarrow \quad compute\ y$
  - Some tricky questions:
    - If the term structure is flat, what is the YTM?
    - If the term structure is flat, and coupon rate = spot rate, what is the price?
    - True/False: We earn a risk-free return **every year** when investing in Treasuries.

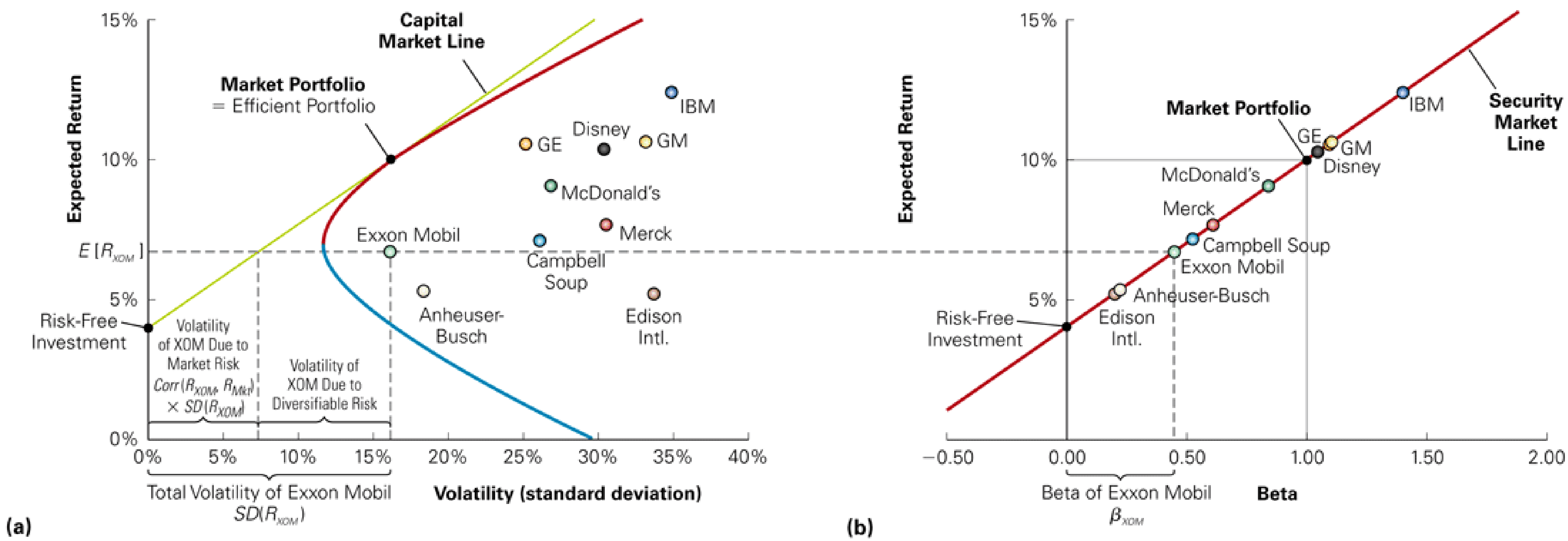
Module 4: Retirement Saving

- For a nominal interest rate r, and inflation i, the real interest rate r_r is:

$$r_r = \frac{1+r}{1+i} - 1$$

- Do some practices:
  - Problem Set 4, Question 5
  - Sample Exam 1, Part II, Question 2

Module 5&6&7: Mean-Variance Analysis



- Minimum variance frontier, Efficient frontier
- CML vs SML
- Mutual fund theorem: All MV optimizers hold a mix of the market (tangency) portfolio and cash.

Module 5&6&7: CAPM

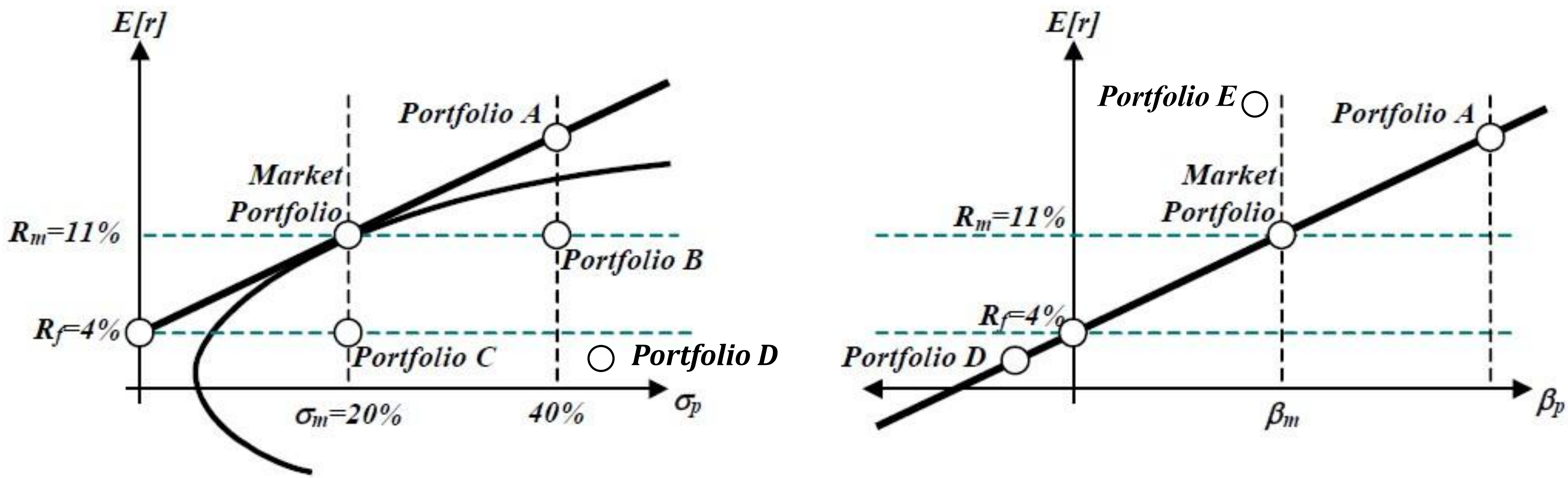
- CAPM regressions:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i\left(R_{m,t} - R_{f,t}\right) + \epsilon_{i,t}$$

  - r-square: 80% of the variation in returns is explained by the variation in the market factor (or is explained by CAPM).
  - $\beta_i\left(R_{m,t} - R_{f,t}\right)$ is the systematic part of asset i's return.
  - $\alpha_i + \epsilon_{i,t}$ is the idiosyncratic part of asset i's return.
  - If CAPM holds, $\alpha_i = 0$.
  - α_i provides a good measure of the risk-adjusted performance for assets.
  - If $\alpha_i > 0$, this asset has extra positive returns. (Underpriced according to CAPM)
  - If $\alpha_i < 0$, this asset has extra negative returns. (Overpriced according to CAPM)

Module 5&6&7: Graph Interpretations of CAPM

- Practice: Assume CAPM holds perfectly



- Construction of A?
- Do you prefer Portfolio B or Portfolio C?
- Market beta of portfolio B, C, D?
- Expected return of Portfolio D?
- Is Portfolio E undervalued or overvalued relative to CAPM?
- Sharpe ratio of E?

Module 8&9: Levered/Unlevered WACC

Market value balance sheet

(Projects owned by the firm + PV of tax shield)	(Claims of capital providers to FCFs from projects/assets)
$\beta_A,\ r_A$ → (Unlevered) Assets	$\beta_E,\ r_E$ → Equity
$\beta_{ts},\ r_{ts}$ → Tax shield $\tau_c D$	$\beta_D,\ r_D$ → Debt

Reflecting business risk, NOT affected by capital structure ← $\beta_A,\ r_A$

- Tax benefit of debt (tax shield): $\tau_c D$

$$V_L = V_U + \tau_c D = D + E$$

  - E.g., if an all-equity firm issues \$100 mil. Debt (tax rate 30%), how much does the firm value increase at the moment of issuing debt?

- Valuation:

$$V_L = \sum_{t=1}^{\infty} \frac{FCF_t}{(1 + WACC_L)^t}, \qquad V_U = \sum_{t=1}^{\infty} \frac{FCF_t}{(1 + r_A)^t} \ (no\ taxes)$$

- Weighted average cost of capital:

$$WACC_L = \frac{D}{V_L}(1 - \tau_c)\, r_D + \frac{E}{V_L}\, r_E, \qquad WACC_U = r_A = \frac{D}{V_L}\, r_D + \frac{E}{V_L}\, r_E \ (no\ taxes)$$

Module 8&9: Steps to Estimate WACC

Step 1. Choose comparable firms (Criterion: similar business risks, similar asset betas)

Step 2: Estimate comparable firms' equity betas (regressions) and debt betas (credit rating)

Step 3: Find the comparable firms' capital structure figures

Step 4: Calculate comparable firms' asset betas **(unlever the equity beta)**:

$$\text{No taxes/Constantly rebalancing debt: } \beta_A = \frac{E}{D+E}\,\beta_E + \frac{D}{D+E}\,\beta_D$$

$$\text{Permanent debt: } \beta_A = \frac{E}{E+(1-\tau_c)D}\,\beta_E + \frac{(1-\tau_c)D}{E+(1-\tau_c)D}\,\beta_D$$

Step 5: Take the average of the comparable firm asset betas to get an estimate of our project's asset beta

Step 6: Calculate project's asset return r_A using CAPM: $r_A = r_f + \beta_A(r_m - r_f)$

This return will be the WACC if the firm is entirely equity-financed. Done.

Step 7: If our project is not entirely equity-financed. Find the capital structure figures for our project. Calculate our project's debt beta and equity beta (**relever the asset beta**):

$$\text{No taxes/Constantly rebalancing debt: } \beta_E = \beta_A + \frac{D}{E}\,(\beta_A - \beta_D)$$

$$\text{Permanent debt: } \beta_E = \beta_A + \frac{D}{E}(1-\tau_c)(\beta_A - \beta_D)$$

Step 8: Calculate our project's equity return, r_E, and debt return, r_D, using CAPM.

Step 9: Calculate the levered WACC of the project: $WACC_L = \frac{D}{D+E}\,(1-\tau_c)\,r_D + \frac{E}{D+E}\,r_E$